

CS768 Midsem Report

Neural Tree-Based Boosted Graph Learning

Pavan Sekhar Naidu Routhu (22B1028)

Siri Bhogi (22B1042)

Saranya Darisipudi (22B1059)

Abstract

Graph Neural Networks (GNNs) have demonstrated strong performance on relational data, while Gradient Boosted Decision Trees (GBDTs) remain highly effective for heterogeneous tabular features. The Boosted Graph Neural Network (BGNN) framework integrates these paradigms by iteratively refining node representations through boosting followed by graph convolution. However, the original BGNN relies on non-differentiable decision trees and limited GNN architectures, restricting optimization flexibility and representational capacity.

This project proposes a redesigned hybrid architecture that replaces classical GBDT modules with differentiable Neural Decision Trees and substitutes standard GCN/GAT backbones with more expressive alternatives such as GraphSAGE, Graph Isomorphism Networks (GIN), and Transformer-based GNNs. The objective is to investigate whether fully differentiable boosting combined with modern graph encoders improves convergence stability, representation learning quality, and predictive performance on benchmark graph datasets.

Introduction

Graph-structured data appears in domains such as citation networks, recommendation systems, biological interaction networks, and social platforms. Learning effective representations from such data requires models that capture both node attributes and relational structure. Graph Neural Networks (GNNs) address this challenge by propagating information across neighboring nodes using message-passing mechanisms. Architectures such as Graph Convolutional Networks (GCNs) and Graph Attention Networks (GATs) have shown strong performance in node-level prediction tasks.

Despite their success, GNNs often struggle when node features are heterogeneous or tabular in nature. In contrast, Gradient Boosted Decision Trees (GBDTs) are highly effective for structured data due to their ability to model complex feature interactions and non-linear relationships. The BGNN framework combines these complementary strengths by iteratively refining node features using boosting before applying graph convolution.

However, the original BGNN framework has two key limitations. First, it relies on non-differentiable decision trees, preventing end-to-end gradient optimization. Second, it

primarily evaluates standard GNN architectures, limiting exploration of more expressive graph encoders. These limitations motivate the proposed redesign.

Problem Formulation

Let a graph be defined as:

$$G = (V, E)$$

where V denotes the set of nodes and E denotes the set of edges. Each node $v_i \in V$ has a feature vector $x_i \in R^d$ and label y_i .

The original BGNN model learns a function:

$$f(X, G) = \text{GNN}(\text{GBDT}(X), G)$$

In this project, we propose the modified formulation:

$$f(X, G) = \text{GNN}_{\theta_g}(\text{NeuralTree}_{\theta_t}(X), G)$$

where:

- $\text{NeuralTree}_{\theta_t}$ is a differentiable tree-based learner,
- GNN_{θ_g} is selected from advanced architectures such as GraphSAGE, GIN, or Transformer-based GNN.

The objectives are:

1. Replace non-differentiable boosting with differentiable neural trees.
2. Compare multiple GNN backbones for representation learning.
3. Evaluate convergence behavior, accuracy, and stability.
4. Analyze the impact of feature refinement on graph prediction tasks.

Literature Review

Graph Neural Networks

Graph Neural Networks generalize deep learning to relational data through neighborhood aggregation. GCN introduced spectral graph convolution, while GAT incorporated attention mechanisms to weigh neighbor contributions. GraphSAGE enabled inductive learning by sampling neighborhoods and learning aggregation functions. GIN improved expressiveness by matching the discriminative power of the Weisfeiler–Lehman graph isomorphism test. More recently, Transformer-based GNNs utilize global attention to capture long-range dependencies.

Gradient Boosted Decision Trees

GBDT is an ensemble learning method that sequentially fits decision trees to residual errors. It is widely used for tabular data due to its robustness and ability to model non-linear feature interactions. However, classical decision trees are non-differentiable and cannot be jointly optimized with neural networks using gradient descent.

Boosted Graph Neural Networks

BGNN integrates boosting and graph learning by iteratively refining node features using decision trees before applying graph convolution. This hybrid approach improves performance when both feature quality and graph structure are important. However, the use of classical trees restricts end-to-end optimization.

Neural Decision Trees

Neural decision trees introduce soft decision boundaries, replacing hard splits with differentiable routing mechanisms. This allows gradient-based optimization and seamless integration with deep learning architectures. Such models combine the interpretability of trees with the flexibility of neural networks.

Research Gap

Existing hybrid graph-boosting frameworks rely on non-differentiable trees and limited GNN architectures. A fully differentiable boosting module combined with modern graph neural architectures remains underexplored. This project aims to address this gap through architectural redesign and empirical evaluation.